

ISP Training Program – AMR DevKit and Perceptor

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2026.08.19



Agenda

- Brief Introduction to AMR
- Overview of AMR DevKit
- Robot Perception
 - AMR DevKit: Bring Up Sensors on Robots
 - AMR Perceptor: Sensing to Reasoning
- Demo
 - AMR Assembly, Bring Up Sensors, Control AMR
 - Run AMR Perceptor

ISP Training Program

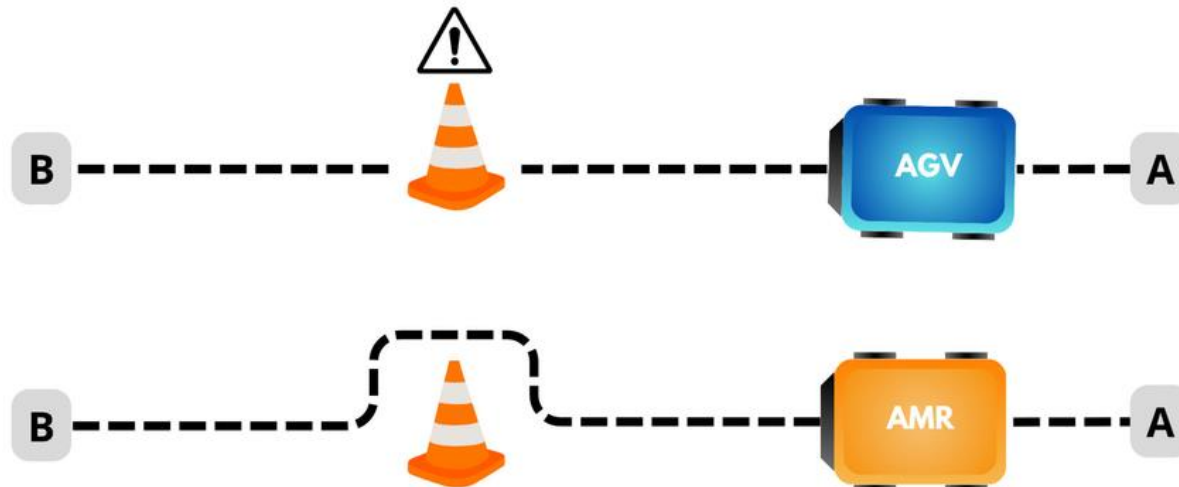
Training Date	Projects	Hardware	NVIDIA Base	GitHub
2026.04.15 (en) 2026.04.22 (zh)	Edge Agent	MIC-733	Agent Studio	LINK
2026.04.29 (en) 2026.05.06 (zh)	VSS on Jetson Thor	MIC-742	VSS 2.4.1	LINK
2026.08.19 (zh)	AMR DevKit AMR Perceptor	AMR DevKit (MIC-732)	-	AMR DevKit Perceptor
TBD	AMR SLAM Navigator	AMR DevKit (MIC-732)	-	LINK
TBD	AMR ReMEmbR	AMR DevKit (MIC-732)	ReMEmbR	ReMEmbR
TBD	Omniverse/Issac Sim	-	-	-

Brief Introduction to AMR



Mobile Robots

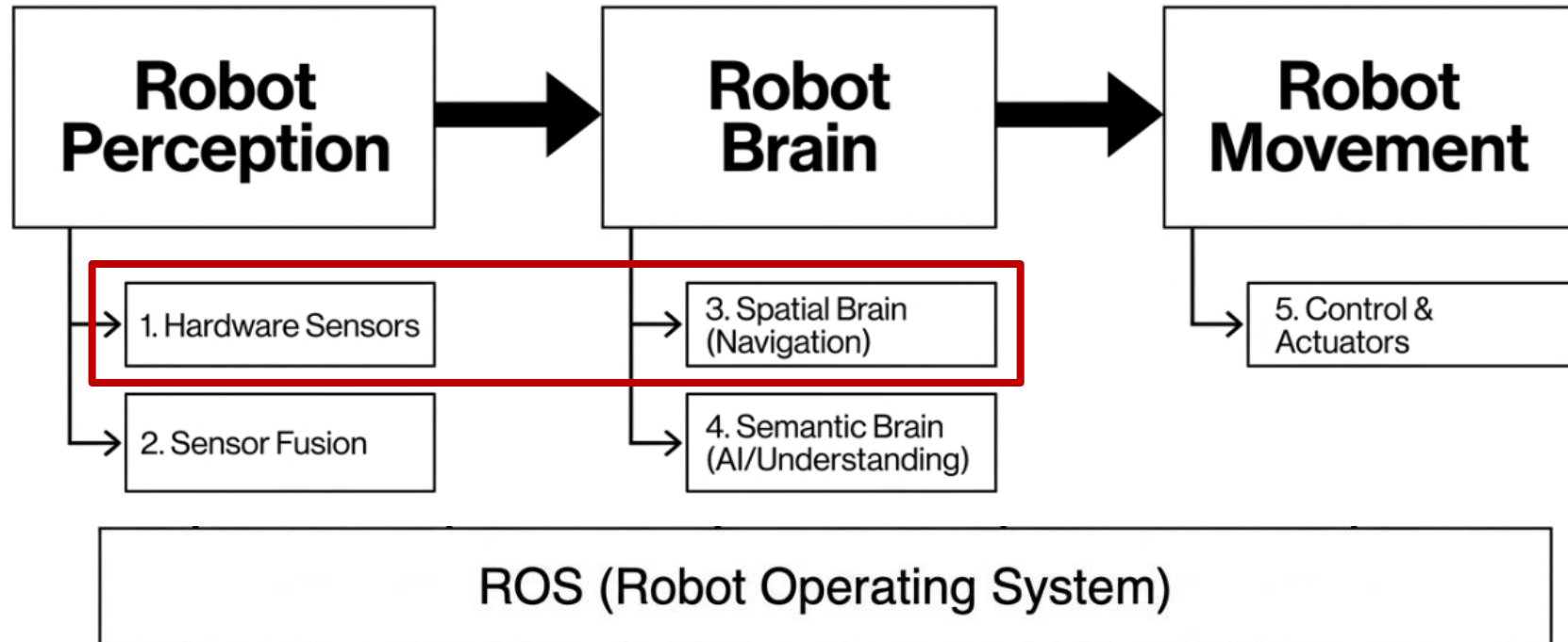
- Common types of mobile robots:
 - Autonomous Guided Vehicles (AGVs)
 - Autonomous Mobile Robots (AMRs)
 - Mobile Manipulator Robots (MMRs)
 - Humanoid Robot (Legged)



AGVs vs. AMRs

- AGVs
 - Follow fixed path and predetermined routes
 - Often require modifying environment (tape, wires, or rails)
 - Follow same path to transport an object
- AMRs
 - Navigation system with mapping, localization abilities
 - Sensors for detecting and avoiding obstacles
 - Path planning

High-level System Design of Robot



For Advantech AMR DevKit,

- It is integrated with all required sensors and be ready for SLAM
- SLAM (Simultaneously Localization and Mapping) will be introduced in upcoming session

Overview of AMR DevKit



Advantech AMR DevKit

- Hardware
 - MIC-732 for computing
 - Sensors (stereo cameras, 2D LiDAR, fisheye cameras)
 - Power supply (battery included and power supply unit for AC)
- Software
 - JetPack 6.0 with ROS package installed
 - Sensor drivers included
 - Support other ISP packages (available on GitHub)
 - AMR Perceptor and AMR ReMEmbR

AMR DevKit: Front Side



AMR DevKit: Rear Side



Light Indicator

- White (blinking): starting
- Light blue (blinking): standby
- Blue: operating
- Yellow: stop operating
- Light orange: low battery
- Red: error

Robot Perception

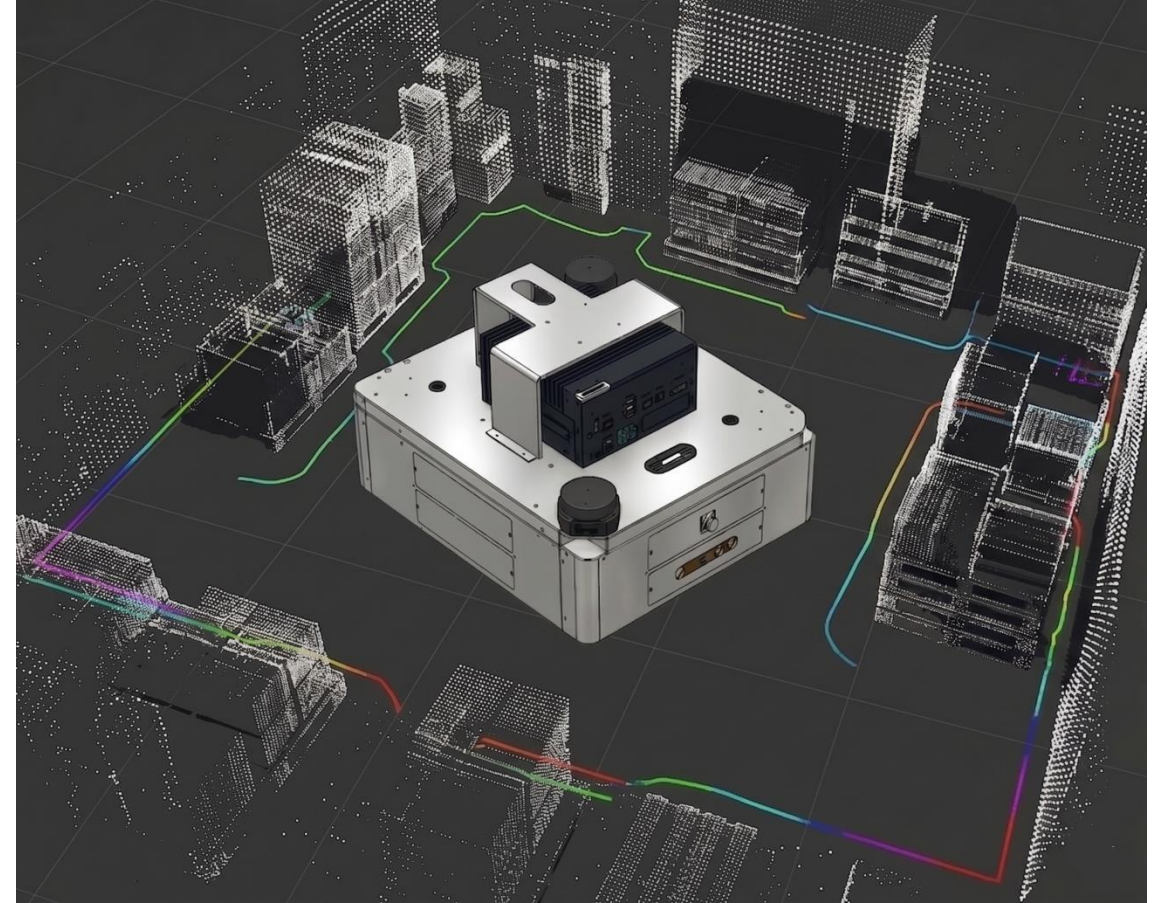


Sensors on AMR DevKit

- Cameras
 - Stereo Camera $\times 2$
 - Measure depth or distance of objects
 - The output is a point cloud
 - Fisheye Camera $\times 2$: not used by AMR DevKit
- 2D LiDAR (Light Detection and Ranging) $\times 2$
 - Two of them are placed diagonally
 - Operate on a single horizontal
 - Emit beams in a fan-like pattern to detect surrounding

Bring Up Sensors

- 3D point cloud of surrounding obstacles, dynamically generated in real-time by stereo cameras
- 2D LiDAR scan profiles mapping the environment boundaries for robot navigation

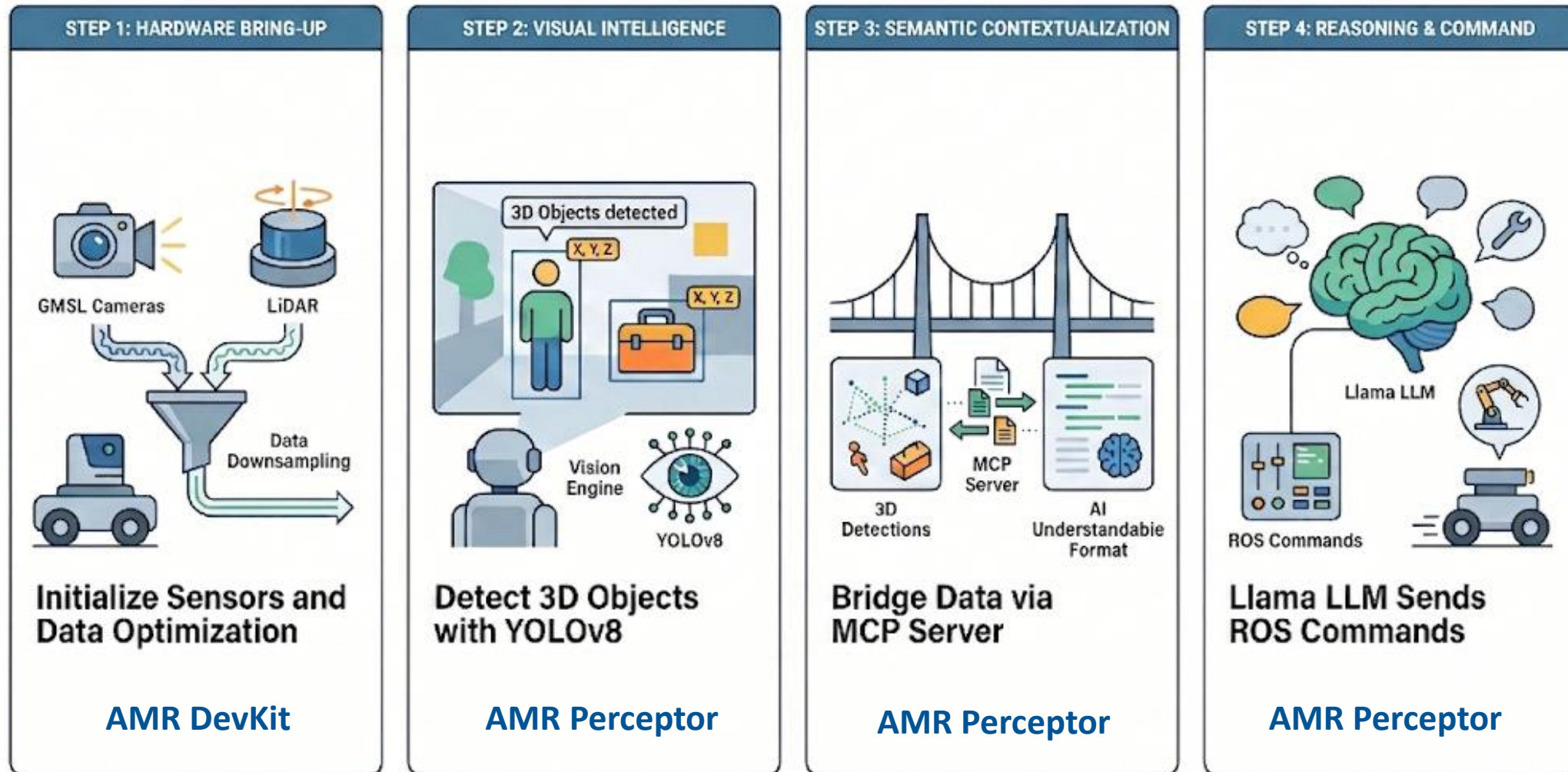


This is a conceptual illustration (AI-generated), not a real screenshot.

AMR Perceptor

- No wheelbase is needed
- Project for demonstrating:
 - Object detection with YOLO: recognizing surrounding objects
 - LLM-driven AMR control via WebUI chat interface

Reasoning Loop



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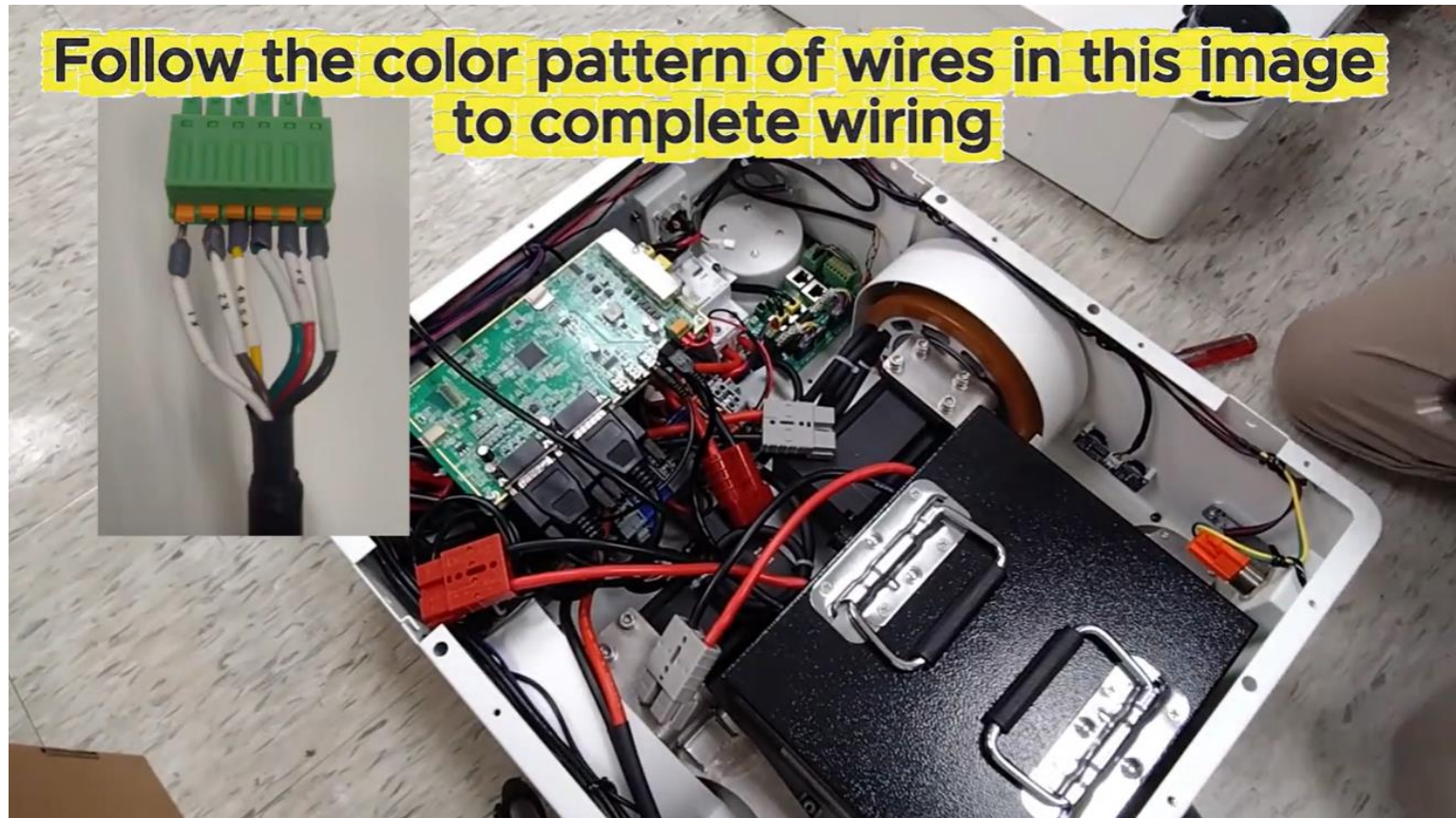
Enabling an Intelligent Planet

Demo



AMR Assembly

- Watch [video tutorial](#)



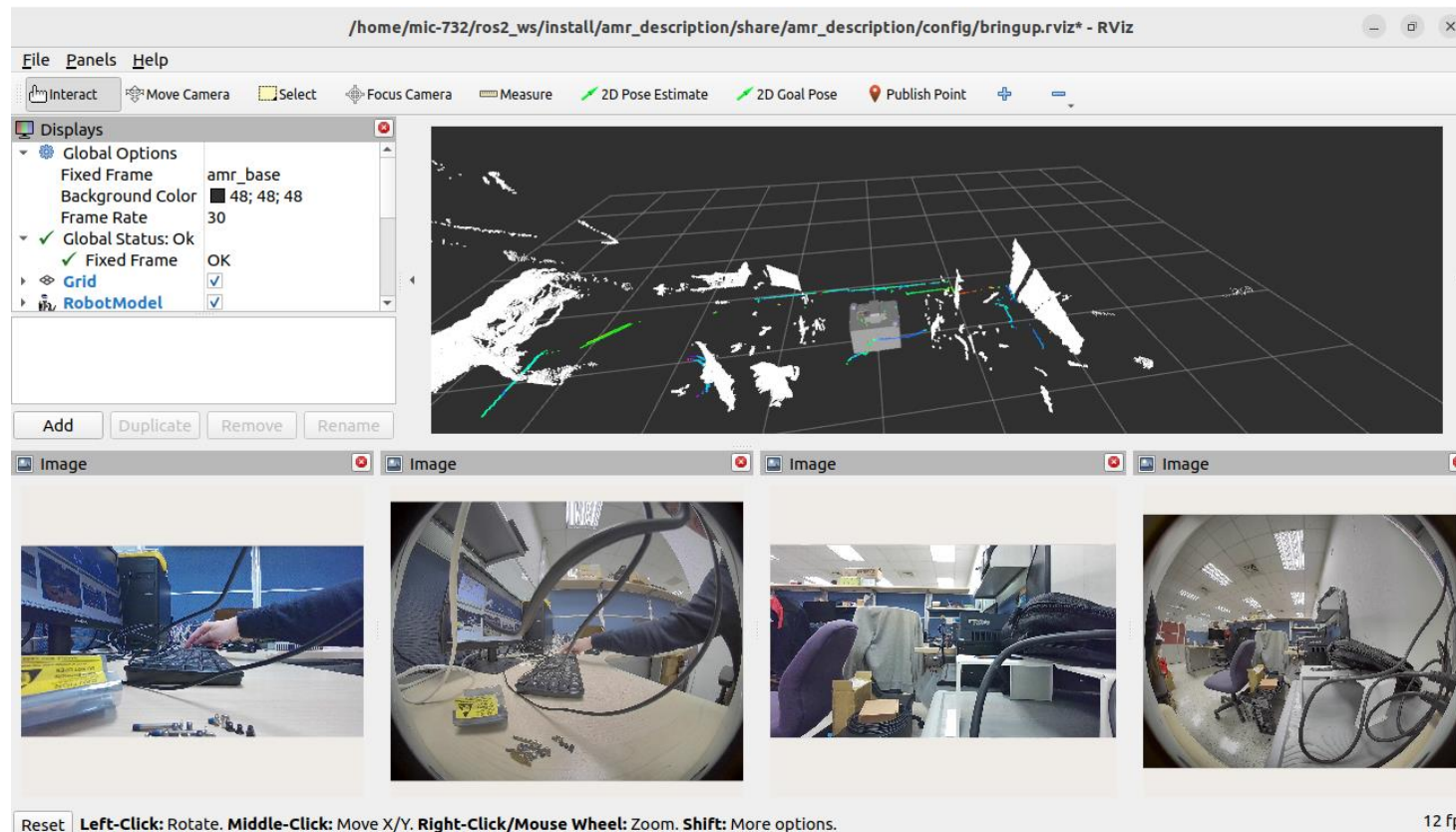
Accessories

- Portable monitor with VESA wall-mounting compatibility
 - Provides live preview of cameras and LiDARs
- PS4 controller
 - Provides direct control for visitors in demo



Bring Up Sensors

- Follow [Installation Guide](#) to install and setup your AMR
- Bring up sensors (watch [video tutorial](#)):
 - `ros2 launch amr_description bringup.launch.py bringup_rviz:=true`



Control AMR with Keyboard

- Connect keyboard with MIC-732 via USB cable
- Enter commands:
 - `cd ~/ros2_amr100`
 - `source docker/scripts/deploy_arm64.sh`
 - `docker exec -u adata -it ros_dev-aarch64-container bash`
 - `source env_setting.sh 99`
 - `ros2 run teleop_twist_keyboard teleop_twist_keyboard`
- Watch [video tutorial](#)
- Key mapping

u (Forward Left)	i (Forward)	o (Forward Right)
j (Rotate Left)	k (Emergency Stop)	l (Rotate Right)
m (Backward Left)	, (Backward)	. (Backward Right)

Control AMR with PS4 Controller

- (Terminal 1)
 - `cd ros2_amr100source docker/scripts/deploy_arm64.sh`
 - `docker exec -u adata -it ros_dev-aarch64-container bash`
 - `source env_setting.sh 99`
 - `ros2 run joy joy_node`
 - Expected outcome:
 - `[INFO] [1786673510.830800078] [joy_node]: No haptic (rumble) available, skipping initialization`
 - `[INFO] [1786673510.831036798] [joy_node]: Opened joystick: PS4 Controller. deadzone: 0.050000`
- (Terminal 2)
 - `docker exec -u adata -it ros_dev-aarch64-container bash`
 - `source env_setting.sh 99ros2 run teleop_twist_joy teleop_node --ros-args \`
 - `-p axis_linear.x:=1 \`
 - `-p scale_linear.x:=0.2 \`
 - `-p axis_angular.yaw:=0 \`
 - `-p scale_angular.yaw:=0.5 \`
 - `-p enable_button:=9`
 - Expected outcome:
 - `[INFO] [1786674169.875979616] [TeleopTwistJoy]: Teleop enable button 9.`
 - `[INFO] [1786674169.876064576] [TeleopTwistJoy]: Linear axis x on 1 at scale 0.200000.`
 - `[INFO] [1786674169.876084928] [TeleopTwistJoy]: Angular axis yaw on 0 at scale 0.500000.`

Control AMR with Mobile App

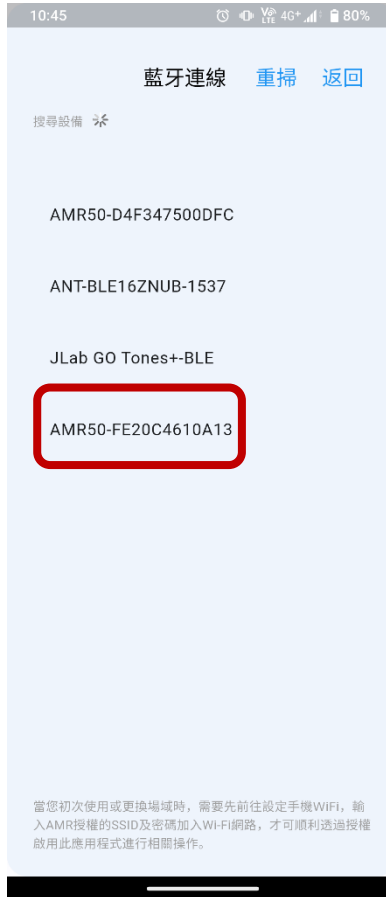
- Chinese interface only
- iOS:
 - Download the app from App Store
- Android:
 - Download the APK from Google Drive and install it manually

Control AMR with Mobile App

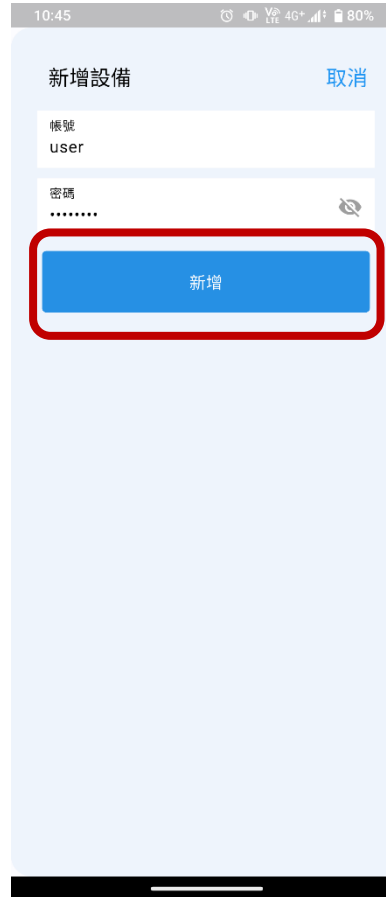
Add AMR



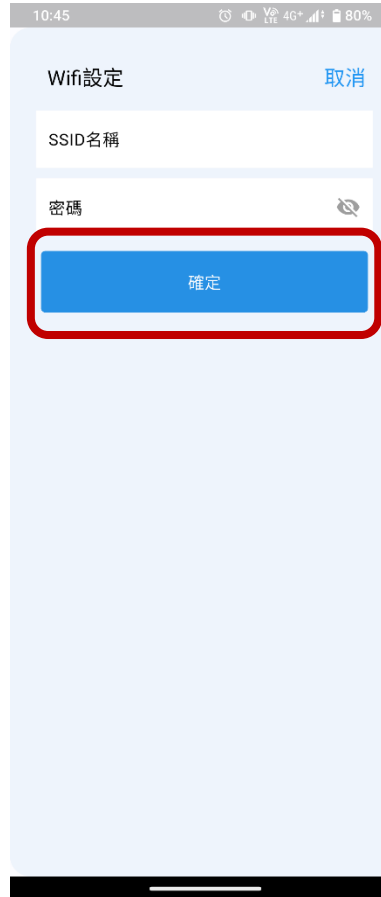
Select your AMR



Continue



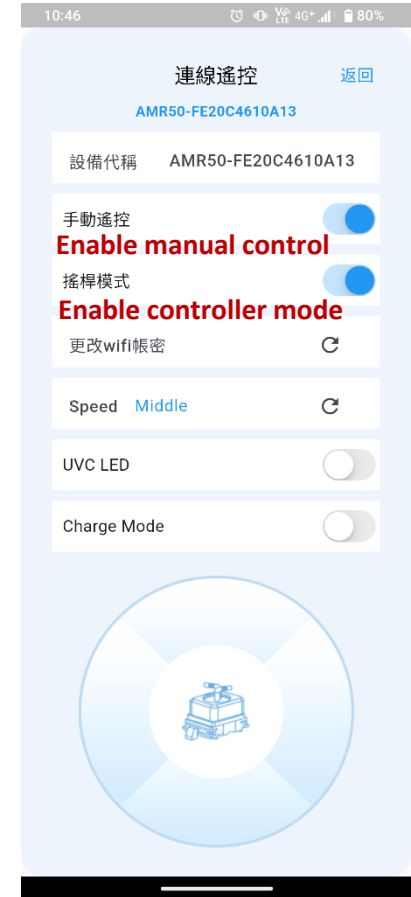
Continue



Connect your AMR



Control your AMR



Run AMR Perceptor

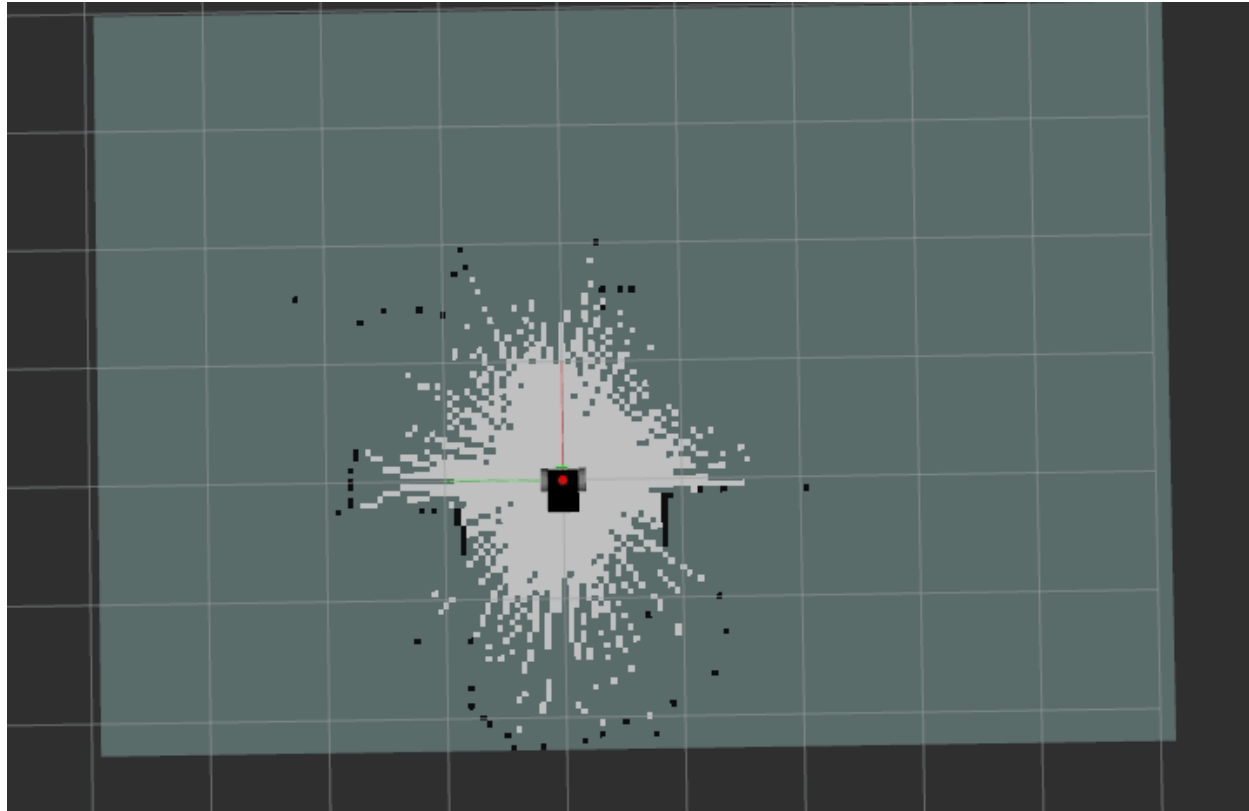
- Start YOLO, MCP, llama.cpp, and Web UI
 - `cd AMR_Perceptor`
 - `docker compose up -d`
- See perception with YOLO detection in Rviz
 - `ros2 launch amr_description bringup.launch.py yolo_rviz:=true preprocess:=true`
- Watch [video tutorial](#)
- For the usage of Web UI, please refer to instructions on GitHub

Follow Instructions on GitHub

- AMR DevKit
 - https://github.com/advantech-EdgeAI/AMR_DevKit
- AMR Perceptor
 - https://github.com/advantech-EdgeAI/AMR_Perceptor
- Any questions? Go to the projects' GitHub Issues

Next Step

- Next time, we will drive AMR to build a map in real world
 - New skill will be unlocked: SLAM



Co-Creating the Future of the IoT World

